The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 December 1993

INCH-POUND

MIL-S-19500/523A
21 July 1993
SUPERSEDING
MIL-S-19500/523(USAF)
30 November 1976

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6383, 2N6384, 2N6385, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500. For JAN product assurance level (see 6.3).

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

	P_T 1/ T_A = +25° C	P_T 2/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{OP} and T_{STG}	$R_{\theta JC}$ Max
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N6383	6.0	100	40	40	5.0	10	0.25	-55 to +175	1.75
2N6384	6.0	100	60	60	5.0	10	0.25	-55 to +175	1.75
2N6385	6.0	100	80	80	5.0	10	0.25	-55 to +175	1.75

1/ Derate linearly 34.2 mW/°C above T_A > +25°C.
2/ Derate linearly 571 mW/°C above T_C > +25°C.

1.4 Primary electrical characteristics.

	h_{FE1}	$V_{CE(SAT)1}$	$V_{CE(SAT)2}$	$V_{BE(ON)1}$	C_{obo}	$\|h_{fe}\|$	Switching	
	V_{CE} = 5.0 V dc I_C = 5.0 A dc 1/	I_C = 5.0 A dc I_B = 10 mA dc 1/	I_C = 10 A dc I_B = 0.1 A dc 1/	V_{CE} = 5.0 V dc I_C = 5.0 A dc 1/	V_{CB} = 10 V dc I_E = 0 100 kHz < f ≤ 1 MHz	V_{CE} = 5.0 V dc I_C = 1.0 A dc f = 1 MHz	V_{CC} = 30 V dc I_{CC} = 5.0 A dc I_{B1} = 20 mA dc	
							t_{on}	t_{off}
		V dc	V dc	V dc	pF		µs	µs
Min	1,000					20		
Max	20,000	2.0	3.0	2.8	200	300	2.5	10.0

1/ Pulsed, see 4.5.1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO) NASA Goddard Space Flight Center, Code 310.A Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A
FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500.

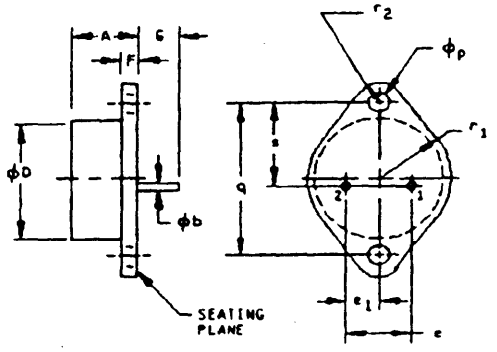
3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
ϕb	.038	.043	0.97	1.09	4
ϕD		.875		22.23	
e	.420	.440	10.67	11.18	
e_1	.205	.225	5.21	5.72	
F	.050	.135	1.27	3.43	
G	.312		7.92		
ϕ_p	.151	.161	3.84	4.09	
q	1.177	1.197	29.90	30.40	
r_1	.495	.525	12.57	13.34	
r_2	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
4. Two leads.
5. Collector shall be electrically connected to the case.

FIGURE 1. Physical dimensions.

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4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels only
11	I_{CEX1} and h_{FE1}
12	See 4.3.1
13	ΔI_{CEX1} = 100 percent of initial value or 100 µA dc, whichever is greater; Δh_{FE1} = ±25 percent of initial value; subgroup 2 of table I herein.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = +162.5°C ±12.5°C; 2N6383 = V_{CB} = 30 V dc;

2N6384 = V_{CB} = 40 V dc; 2N6385 = V_{CB} = 60 V dc.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

a. Subgroup 3: Intermittent operation life; method 1037; $V_{CE} \geq 10$ V dc, ΔT_J between cycles = 100°C.

b. Subgroup 5: Thermal resistance; method 3131; V_{CE} = 20 V dc; I_C = 10 A; $R_{\theta JC}$ = 1.75°C/W maximum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Terminal strength (tension); method 2036; test condition A; weight = 10 pounds; t = 15 s.

b. Subgroup 6: Intermittent operation life; method 1037; $V_{CE} \geq 10$ V dc, ΔT_J between cycles = 100°C.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D I_C = 200 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6383				40		V dc
2N6384				60		V dc
2N6385				80		V dc
Collector to emitter breakdown voltage	3011	Bias condition B I_C = 200 mA dc R_{BB} = 100Ω pulsed (see 4.5.1)	$V_{(BR)CER}$			
2N6383				40		V dc
2N6384				60		V dc
2N6385				80		V dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}		1.0	mA dc
2N6383		V_{CE} = 40 V dc				
2N6384		V_{CE} = 60 V dc				
2N6385		V_{CE} = 80 V dc				
Emitter to base cutoff current	3061	Bias condition D V_{EB} = 5.0 V dc	I_{EBO}		5.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A V_{BE} = 1.5 V dc	I_{CEX1}		0.3	mA dc
2N6383		V_{CE} = 40 V dc				
2N6384		V_{CE} = 60 V dc				
2N6385		V_{CE} = 80 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		1.0	mA dc
2N6383		V_{CE} = 40 V dc				
2N6384		V_{CE} = 60 V dc				
2N6385		V_{CE} = 80 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base emitter voltage (nonsaturated)	3066	Test condition B V_{CE} = 3.0 V dc I_C = 5.0 A dc	$V_{BE(on)1}$		2.8	V dc
Base emitter voltage (nonsaturated)	3066	Test condition B V_{CE} = 3.0 V dc I_C = 10 A dc pulsed (see 4.5.1)	$V_{BE(on)2}$		4.5	V dc
Saturated voltage and resistance	3071	I_C = 5.0 A dc I_B = 10 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		2.0	V dc
Saturated voltage and resistance	3071	I_C = 10 A dc I_B = 0.1 A dc pulsed (see 4.5.1)	$V_{CE(sat)2}$		3.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 3 V dc I_C = 5 A dc pulsed (see 4.5.1)	h_{FE1}	1000	20,000	
Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc I_C = 10 A dc pulsed (see 4.5.1)	h_{FE2}	100		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A V_{BE} = 1.5 V dc	I_{CEX2}		3.0	mA dc
2N6383 2N6384 2N6385		V_{CE} = 40 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc I_C = 5.0 A dc pulsed (see 4.5.1)	h_{FE3}	200		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Pulse response	3251	Test Condition A, except test circuit and pulse requirements in accordance with figure 2				
Turn-on time		V_{CC} = 30 V dc I_C = 5.0 A dc I_{B1} = 20 mA dc	t_{on}		2.5	µs
Turn-off time		V_{CC} = 30 V dc I_C = 5.0 A dc $I_{B1} = -I_{B2}$ = 20 mA dc	t_{off}		10.0	µs
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc I_C = 1 A dc f = 1.0 MHz	$\|h_{fe}\|$	20	300	
Open circuit output capacitance	3236	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	C_{obo}		200	pF
Subgroup 5						
Safe operating area (continuous dc)	3053	T_C = +25°C t = 1.0 s, 1 cycle See figure 3				
Test 1 (Both device types)		V_{CE} = 10 V dc I_C = 10 A dc				
Test 2 (All device types)		V_{CE} = 30 V dc I_C = 3.33 A dc				
Test 3 2N6383		V_{CE} = 40 V dc I_C = 1.5 A dc				
2N6384		V_{CE} = 60 V dc I_C = 0.4 A dc				
2N6385		V_{CE} = 80 V dc I_C = 0.16 A dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Electrical measurements		See table II, steps 1 and 3				
Safe operating area (switching)	3053	Load condition C (unclamped inductive load) See figure 4 T_C = 25°C duty cycle ≤ 10% R_S = 0.1Ω				
Test 1		t_p approx 1 ms (vary to obtain I_C) R_{B81} = 1 kΩ V_{B81} = 10 V dc R_{B82} = ∞ V_{B82} = 0 V V_{CC} = 30 V dc I_C = 10 A dc R_L ≤ 0.5Ω L = 1 mH at 10 A dc				
Electrical measurements		See table II, steps 1 and 3				
Test 2		t_p approximately 1 ms (vary to obtain I_C) R_{B81} = 10 kΩ V_{B81} = 10 V dc R_{B82} = ∞ V_{B82} = 0 V V_{CC} = 30 V dc I_C = 0.2 A dc L = 100 mH at 0.2 A dc R_L ≤ 0.5Ω				
Safe operating area (switching)	3053	Load condition B (clamped inductive load) See figure 5 T_A = 25°C $t_r + t_f$ ≤ 1.0 µs duty cycle ≤ 10% t_p = 5 ms (vary to obtain I_C) R_s = 0.1Ω V_{CC} = 10 V dc I_C = 10 A dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	-
Subgroup 5 - Continued						
2N6383		Clamp voltage = 40 V dc				
2N6384		Clamp voltage = 60 V dc				
2N6385		Clamp voltage = 80 V dc				
		Device fails if clamp voltage is not reached				
Electrical measurements		See table II, steps 1 and 3				

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current 2N6383 2N6384 2N6385	3041	Bias condition A V_{BE} = 1.5 V dc V_{CE} = 40 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX1}		0.3	mA dc
2.	Collector to emitter cutoff current 2N6383 2N6384 2N6385	3041	Bias condition A V_{BE} = 1.5 V dc V_{CE} = 40 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX3}		0.6	mA dc
3.	Forward-current transfer ratio	3076	V_{CE} = 3.0V dc I_C = 5.0 A dc pulsed (see 4.5.1)	h_{FE1}	1000	20,000	
4.	Forward-current transfer ratio	3076	V_{CE} = 3.0 V dc I_C = 5.0 A dc pulsed (see 4.5.1)	h_{FE4}	800	24,000	
5.	Thermal Response	3161		V_{BE}			3/

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows.

a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroup 3, see table II herein, steps 2, 4, and 5.

c. Subgroup 6, see table II herein, steps 2 and 4.

2/ The electrical measurements for table V of MIL-S-19500 are as follows.

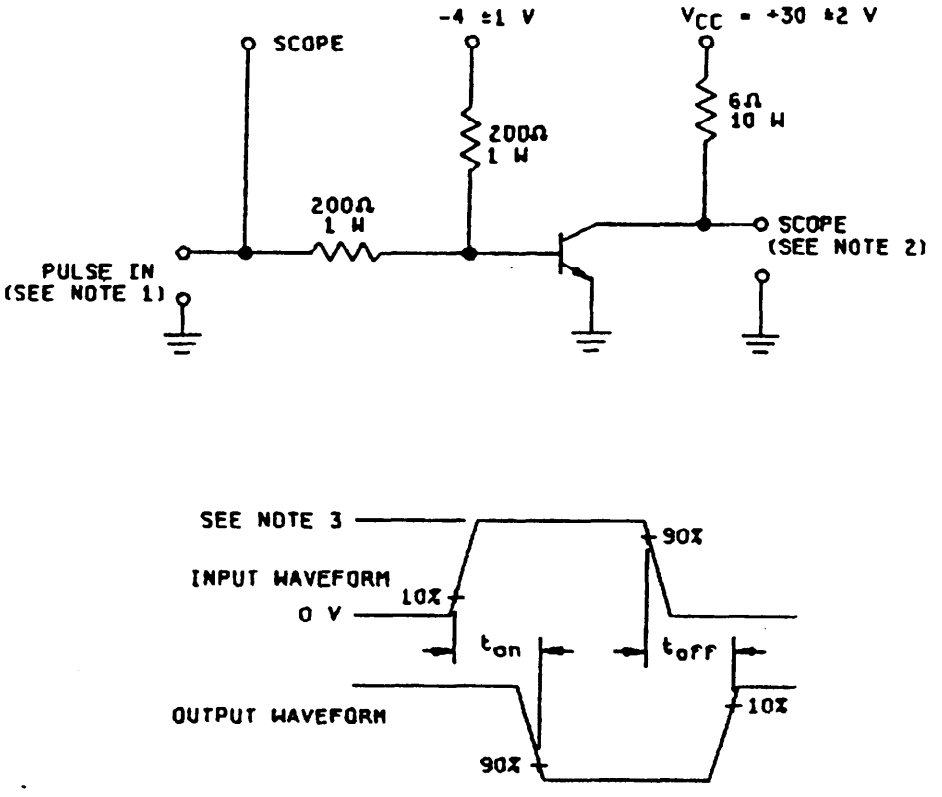
a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroup 3, see table II herein, steps 1 and 3.

c. Subgroup 6, see table II herein, steps 2, 4, and 5.

3/ Twenty percent degradation in group B is permitted; thirty percent degradation in group C is permitted.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each < 20 nanoseconds; duty cycle < 2 percent, generator source impedance shall be 50Ω; pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{IN} > 100 kΩ; C_{IN} < 50 pf; rise time < 2 nanoseconds.
3. Pulse In shall be 10 V maximum.

FIGURE 2. Pulse response test circuit.

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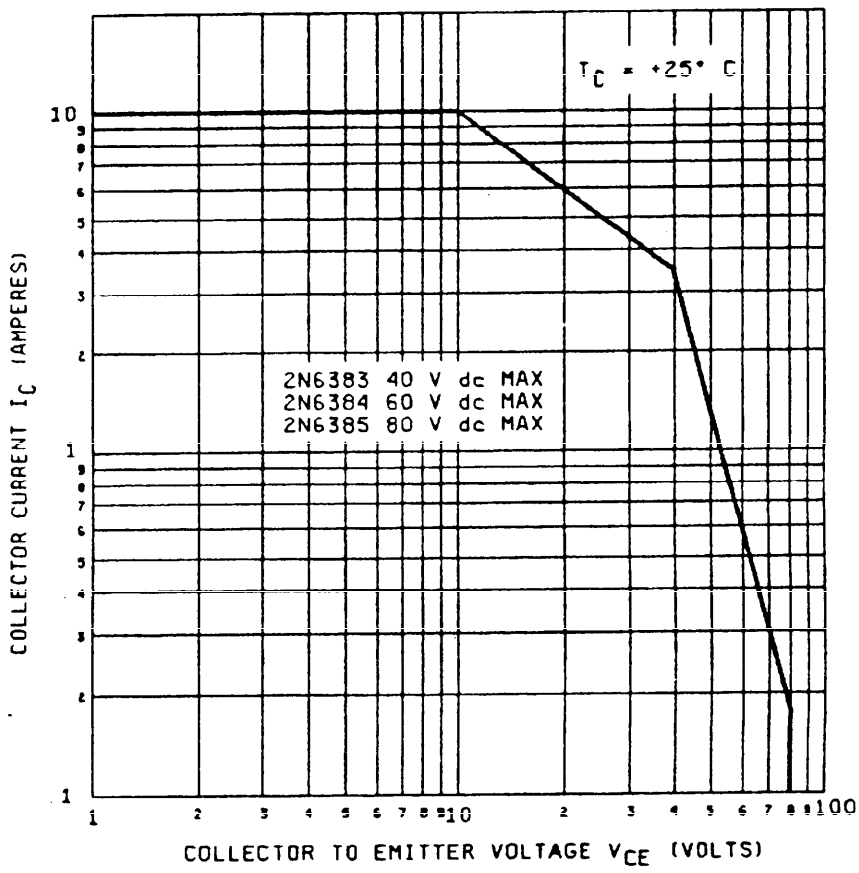


FIGURE 3. Maximum safe operating graph (continuous dc).

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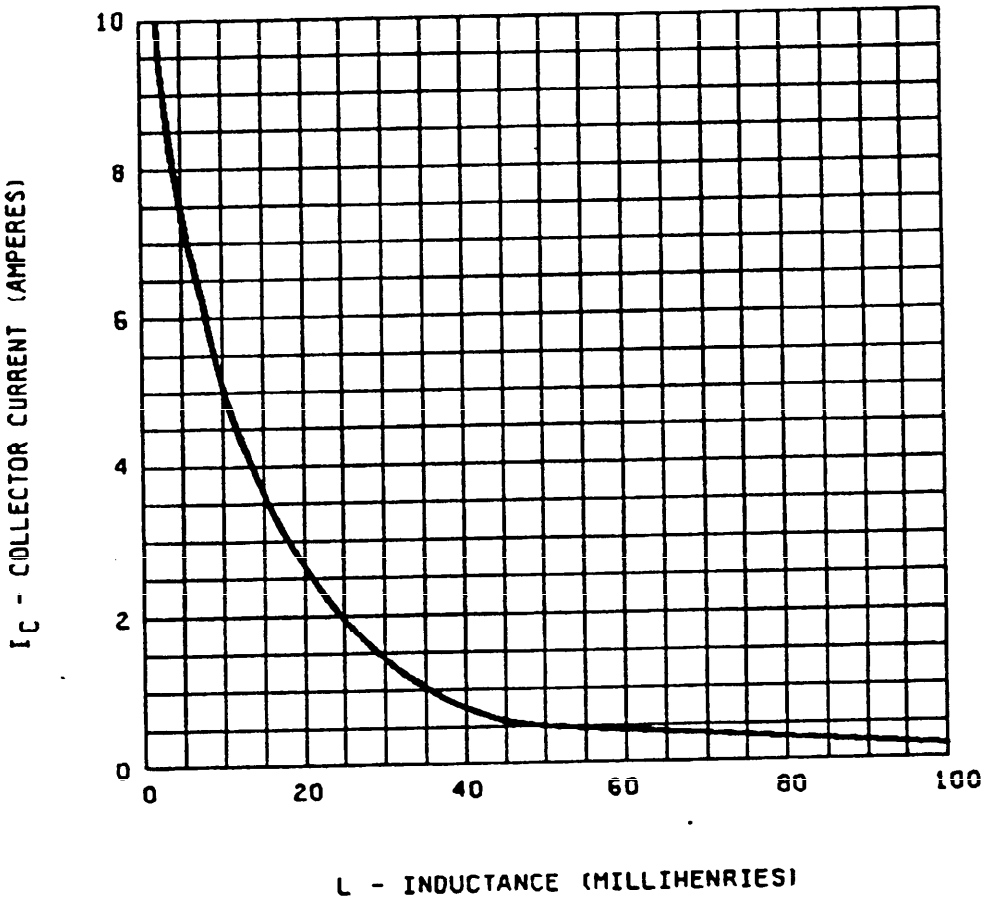
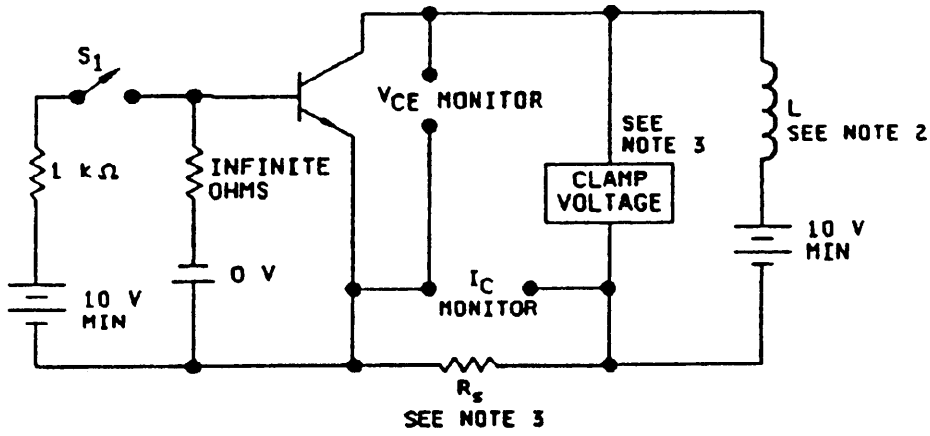


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 1 mH at 10 A with a max dc resistance of .1 ohm.
3. $R_S \leq .1$ oh. 12 W, 1% tolerance max (noninductive).
4. With switch S1 closed, set the specified test conditions.
5. Open S1. Device fails if clamp voltage is not reached and maintained until the current returns to zero.
6. Perform specified end point tests.

FIGURE 5. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type number and quality assurance (JAN) level.

6.3 Replacement data. JANTX devices are oneway replacements for JAN devices (example JANTX2N6383 for JAN2N6383).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Air Force - 19, 85, 99

User activities:
Air Force - 13, 15

Preparing activity:
NASA - NA

Agent:
DLA - ES

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/523A	2. DOCUMENT DATE (YYMMDD) 930721

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE DARLINGTON TRANSISTOR, NPN, SILICON, POWER TYPES 2N6383 2N6384, 2N6385, JANTX AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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